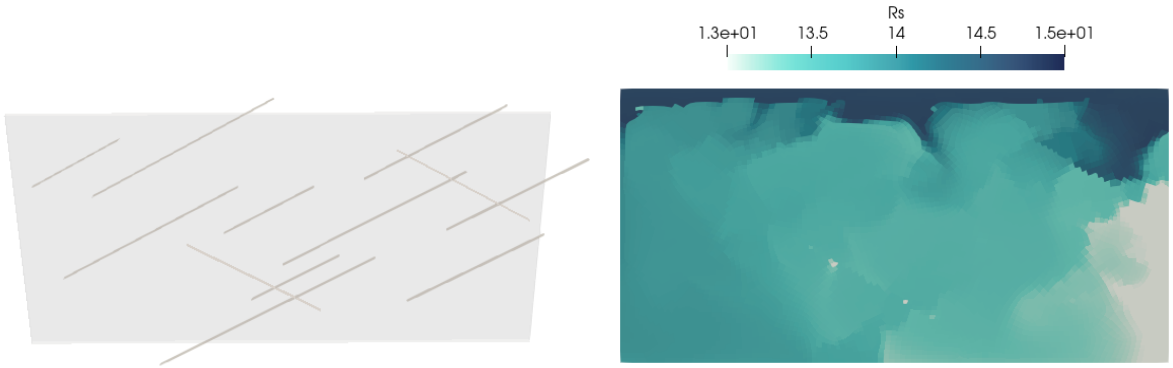
DARSim Example:

Fractures (pEDFM) 2D CCS



This example demonstrates how to use pEDFM capabilities to simulate a fractured reservoir. Here, CO_2 is injected in the bottom left. The figure above shows the locations of the fractures (left) and the resulting dissolved CO_2 distribution after approximately 125 years.

The fracture properties and locations of the fractures are defined in `FractureInput.txt` along with parameters controlling the (p)EDFM methodology. For Cartesian and corner-point grids, the fracture generator must first be run with:

```
>> DARSimFracGen("path/to/FractureInput","FractureInput.txt")
```

This generates the fracture output, called `Fracture_Output.txt`. This file is then passed to `Sim.txt` with:

```
--Keyword for fractures
FRACTURED
      ON
      Fracture_Output.txt
```

For unstructured mesh types, the fracture generation and simulation are completed simultaneously when DARSim is called in Matlab. In this case, the `Sim.txt` file defines the fracture file as:

```
--Keyword for fractures
FRACTURED
      ON
      FractureInput.txt
```